

Machine Learning: Regularization

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1 Why Move Beyond Ordinary Least Squares?

Consider the linear regression model

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon. \quad (1)$$

Ordinary least squares (OLS) is attractive because it is simple, familiar, and easy to interpret. When the relationship is approximately linear and the number of observations is large relative to the number of predictors ($n \gg p$), OLS often performs well. In this setting, the coefficient estimates tend to be stable and have relatively low variance.

As p becomes large relative to n , however, two problems can arise.

- If p is close to n , or if the predictors are strongly correlated, the regression problem may become poorly conditioned. Small changes in the estimation sample can then produce large changes in the OLS coefficients. A model may consequently fit the estimation sample very well while predicting new observations poorly.
- If $p > n$, the OLS coefficients are not uniquely determined. There are more coefficients than observations, so multiple coefficient vectors can produce the same fitted values in the estimation sample while generating very different predictions for new observations.

These concerns motivate procedures designed to improve two features central to machine learning: **prediction accuracy** and **model interpretability**. Regularization addresses both by restricting or shrinking the coefficient estimates.

Model interpretability. When many predictors are included, some may contribute little to prediction. A simpler model can be easier to understand and communicate. Three broad approaches are common:

Approach	Main idea
Subset selection	Select some of the original predictors and fit OLS using only those variables.
Shrinkage	Retain the predictors but penalize large coefficients. Ridge, lasso, and elastic net are the main examples.
Dimension reduction	Replace the original predictors with a smaller number of constructed combinations or components.

This note focuses on the shrinkage methods: **ridge regression**, the **lasso**, and the **elastic net**.

2 Shrinkage and Regularization

2.1 Core Idea

OLS searches over all possible coefficient vectors and selects the one that minimizes the residual sum of squares (RSS):

$$\text{RSS}(\boldsymbol{\beta}) = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2.$$

Shrinkage methods instead minimize RSS subject to a restriction on the overall size of the slope coefficients. Equivalently, they solve

$$\min_{\beta_0, \boldsymbol{\beta}} \{ \text{RSS}(\boldsymbol{\beta}) + \lambda P(\boldsymbol{\beta}) \}, \quad (2)$$

where $P(\boldsymbol{\beta})$ measures coefficient size and $\lambda \geq 0$ controls the strength of regularization. The intercept is normally not penalized.

Two forces therefore operate simultaneously:

- **Fit the data:** choose coefficients that make RSS small.
- **Control complexity:** discourage coefficients from becoming unnecessarily large.

When $\lambda = 0$, the solution is OLS. As λ increases, the penalty receives greater weight and the coefficients are shrunk more strongly toward zero.

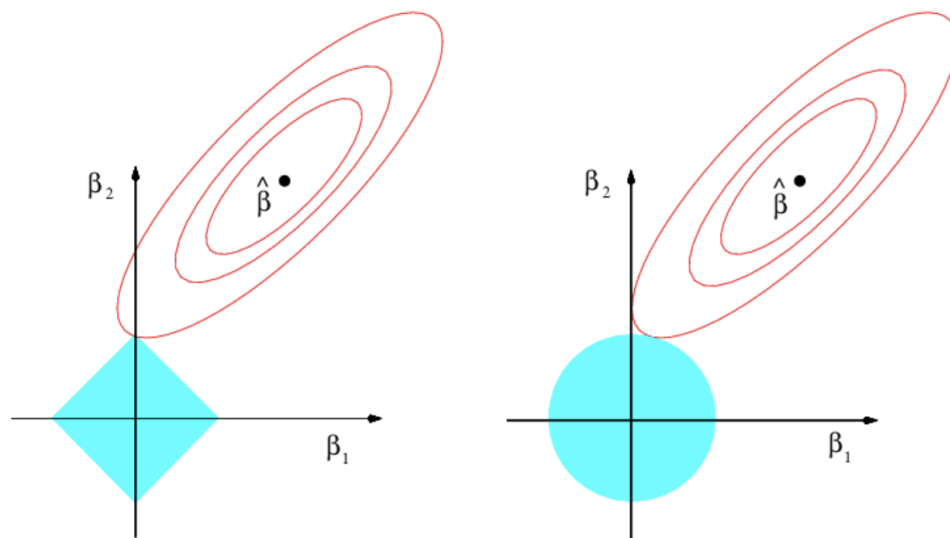


Figure 1: RSS contours and constraint regions: lasso (left) and ridge (right)

Figure 1 illustrates the geometry of regularization. The red ellipses are RSS contours: every coefficient vector on a given contour produces the same RSS, and contours closer to the center represent a better fit to the estimation sample. The center is the unconstrained OLS estimate.

The sky-blue regions contain the coefficient values that the regularized model is permitted to choose. In two dimensions, the lasso constraint is

$$\|\beta\|_1 = |\beta_1| + |\beta_2| \leq s,$$

which produces a diamond. The ridge constraint is

$$\|\beta\|_2 = \sqrt{\beta_1^2 + \beta_2^2} \leq s,$$

which produces a circle. A smaller s corresponds to stronger regularization. The regularized estimate is the point at which the lowest attainable RSS contour first touches the feasible region.

The different shapes also explain why the lasso can produce exact zero coefficients. The lasso region has corners on the coordinate axes, so the first point of contact can occur at a corner. The ridge boundary is smooth, so ridge typically shrinks coefficients toward zero without setting them exactly equal to zero.

2.2 Ridge Regression

Ridge regression solves

$$\min_{\beta_0, \beta_1, \dots, \beta_p} \left\{ \text{RSS} + \lambda \sum_{j=1}^p \beta_j^2 \right\}. \quad (3)$$

It uses the squared ℓ_2 penalty. A larger λ produces stronger shrinkage, and the slope coefficients approach zero as λ becomes very large. Ridge can substantially stabilize estimates when predictors are numerous or highly correlated. Its main limitation is that it generally does not set coefficients exactly equal to zero and therefore does not produce a sparse model.

Standardization. The penalty operates on the numerical sizes of the coefficients. If predictors are measured on different scales, the same penalty does not treat them comparably. Predictors should therefore generally be standardized before ridge, lasso, or elastic-net estimation.

2.3 The Least Absolute Shrinkage and Selection Operator (LASSO)

The lasso replaces the squared penalty with an absolute-value penalty:

$$\min_{\beta_0, \beta_1, \dots, \beta_p} \left\{ \text{RSS} + \lambda \sum_{j=1}^p |\beta_j| \right\}. \quad (4)$$

This ℓ_1 penalty can set some coefficient estimates exactly equal to zero. The lasso therefore performs shrinkage and variable selection simultaneously. A model with many zero coefficients is called **sparse**; sparsity can make a fitted model easier to interpret when the original predictor set is large.

2.4 Elastic Net

Elastic net combines the lasso and ridge penalties:

$$\min_{\beta_0, \beta_1, \dots, \beta_p} \left\{ \text{RSS} + \lambda \left[\alpha \sum_{j=1}^p |\beta_j| + (1 - \alpha) \sum_{j=1}^p \beta_j^2 \right] \right\}, \quad (5)$$

where $\lambda \geq 0$ controls the overall strength of regularization and $\alpha \in [0, 1]$ controls the mixture of the two penalties:

- $\alpha = 1$: elastic net becomes the lasso;

- $\alpha = 0$: elastic net becomes ridge regression;
- $0 < \alpha < 1$: both ℓ_1 and ℓ_2 penalties are used.

Elastic net is particularly useful when predictors are highly correlated. The lasso may select one predictor from a correlated group and exclude the others, and that choice can change across estimation samples. The ridge component encourages more stable treatment of correlated predictors, while the lasso component preserves the possibility of exact zeros. Elastic net can therefore combine the stability of ridge with the sparsity of the lasso. This feature is useful in macroeconomic applications, where several variables may measure closely related aspects of economic activity.

2.5 Comparison

	Ridge	LASSO	Elastic Net
Penalty	ℓ_2	ℓ_1	Mixture of ℓ_1 and ℓ_2
Main effect	Shrinks coefficients	Shrinks and can set coefficients to zero	Shrinks, selects, and stabilizes correlated predictors
Variable selection	Typically no	Yes	Yes, when $\alpha > 0$
Typical strength	Many predictors with small effects	A sparse underlying signal	Sparse prediction with correlated predictors
Tuning parameters	λ	λ	λ and α

No method dominates in every application. Ridge can work well when many predictors contain small amounts of useful information. The lasso is appealing when a small subset is expected to contain most of the predictive signal. Elastic net is a useful compromise when sparsity is desirable but predictors are strongly correlated.

3 The Bias–Variance Tradeoff

Shrinkage deliberately introduces some bias but can substantially reduce variance. To see why this tradeoff can improve prediction, begin by defining the criterion used to evaluate a fitted model.

3.1 Out-of-Sample Mean Squared Error

An **out-of-sample prediction** is a prediction for an observation that was not used to estimate the model. For a separate test sample, prediction performance can be measured by

$$\text{TestMSE} = \frac{1}{n_{\text{test}}} \sum_{i \in \mathcal{I}_{\text{test}}} [Y_i - \hat{f}(X_i)]^2. \quad (6)$$

A smaller test MSE indicates more accurate predictions for new observations.

The theoretical counterpart is the expected squared prediction error at a new predictor value x . Let $\mathcal{D}$ denote the estimation sample and write $\hat{f}_{\mathcal{D}}(x)$ to emphasize that the fitted prediction depends on that sample. Then

$$\text{MSE}(x) = \mathbb{E}_{\mathcal{D}, \varepsilon} \left[\left\{ Y_{\text{new}} - \hat{f}_{\mathcal{D}}(x) \right\}^2 \middle| X_{\text{new}} = x \right]. \quad (7)$$

The expectation averages over both the new outcome and the estimation samples that could have been observed.

3.2 Decomposition of MSE

Suppose

$$Y_{\text{new}} = f(x) + \varepsilon_{\text{new}}, \quad \mathbb{E}[\varepsilon_{\text{new}} \mid X_{\text{new}} = x] = 0, \quad \text{Var}(\varepsilon_{\text{new}} \mid X_{\text{new}} = x) = \sigma^2.$$

Define the average fitted prediction across repeated estimation samples as

$$\bar{f}(x) = \mathbb{E}_{\mathcal{D}} [\hat{f}_{\mathcal{D}}(x)].$$

Step 1: Separate future noise from estimation error. Substituting $Y_{\text{new}} = f(x) + \varepsilon_{\text{new}}$ into Equation (7) and expanding the square gives

$$\begin{aligned} \text{MSE}(x) &= \mathbb{E}_{\mathcal{D}} \left[\left\{ f(x) - \hat{f}_{\mathcal{D}}(x) \right\}^2 \right] + \mathbb{E} [\varepsilon_{\text{new}}^2] \\ &\quad + 2\mathbb{E}_{\mathcal{D}, \varepsilon} [\varepsilon_{\text{new}} \{f(x) - \hat{f}_{\mathcal{D}}(x)\}]. \end{aligned} \quad (8)$$

The final cross-product equals zero because the new error has conditional mean zero and is independent of the estimation sample. Because $\mathbb{E}[\varepsilon_{\text{new}}^2] = \sigma^2$, Equation (8) becomes

$$\text{MSE}(x) = \sigma^2 + \mathbb{E}_{\mathcal{D}} \left[\left\{ f(x) - \hat{f}_{\mathcal{D}}(x) \right\}^2 \right]. \quad (9)$$

Step 2: Separate bias from variance. Add and subtract $\bar{f}(x)$:

$$f(x) - \hat{f}_{\mathcal{D}}(x) = \{f(x) - \bar{f}(x)\} + \{\bar{f}(x) - \hat{f}_{\mathcal{D}}(x)\}.$$

Squaring and averaging over estimation samples yields

$$\begin{aligned} \mathbb{E}_{\mathcal{D}} \left[\{f(x) - \hat{f}_{\mathcal{D}}(x)\}^2 \right] \\ = \{f(x) - \bar{f}(x)\}^2 + \mathbb{E}_{\mathcal{D}} \left[\{\hat{f}_{\mathcal{D}}(x) - \bar{f}(x)\}^2 \right] \\ + 2 \{f(x) - \bar{f}(x)\} \mathbb{E}_{\mathcal{D}} [\bar{f}(x) - \hat{f}_{\mathcal{D}}(x)]. \end{aligned}$$

The last term is zero because

$$\mathbb{E}_{\mathcal{D}} [\bar{f}(x) - \hat{f}_{\mathcal{D}}(x)] = 0.$$

The first remaining term is squared bias, and the second is variance. Substituting them into Equation (9) gives

$$\boxed{\text{MSE}(x) = \underbrace{\sigma^2}_{\text{irreducible noise}} + \underbrace{\{\mathbb{E}_{\mathcal{D}}[\hat{f}_{\mathcal{D}}(x)] - f(x)\}^2}_{\text{squared bias}} + \underbrace{\text{Var}_{\mathcal{D}}[\hat{f}_{\mathcal{D}}(x)]}_{\text{variance}}}. \quad (10)$$

Connecting regularization, the bias–variance tradeoff, and MSE. The bias–variance tradeoff arises from how the estimator changes as the regularization constraint becomes stronger. Figure 1 provides the geometric intuition. The figure represents one estimation sample, but across repeated estimation samples the center of the red RSS contours—the OLS estimate—can move. When p is large relative to n , this movement may be substantial, producing high variance.

The sky-blue constraint restricts how closely the regularized estimate can follow these sample-specific movements. A stronger constraint therefore typically reduces variance by making the estimate more stable across estimation samples. At the same time, it pulls the estimate away from the unrestricted OLS solution, generally toward zero, and thereby introduces bias. This opposing movement—higher squared bias together with lower variance—is the bias–variance tradeoff.

The MSE provides a criterion for evaluating its consequence for prediction:

$$\text{MSE} = \sigma^2 + \text{Bias}^2 + \text{Variance}.$$

For example, suppose $\sigma^2 = 2$ and, under a correctly specified model, OLS and two regularized

estimators have the following properties:

	Irreducible noise	Squared bias	Variance	MSE
OLS	2	0	10	12
Moderate regularization	2	2	4	8
Strong regularization	2	8	1	11

Relative to OLS, moderate regularization increases squared bias by 2 but reduces variance by 6, so MSE falls from 12 to 8. Strong regularization reduces variance further, but the additional squared bias is sufficiently large to raise MSE to 11.

Thus, a bias–variance tradeoff means that stronger regularization generally improves one component while worsening the other. MSE measures the net effect, and the preferred amount of regularization is the one that produces the lowest out-of-sample MSE.

4 Remarks

4.1 Penalized and Constrained Forms

The three shrinkage methods can also be written as constrained optimization problems:

$$\textbf{Ridge:} \quad \min_{\beta_0, \beta} \text{RSS} \quad \text{subject to} \quad \|\beta\|_2 \leq s,$$

$$\textbf{Lasso:} \quad \min_{\beta_0, \beta} \text{RSS} \quad \text{subject to} \quad \|\beta\|_1 \leq s,$$

$$\textbf{Elastic net:} \quad \min_{\beta_0, \beta} \text{RSS} \quad \text{subject to} \quad \alpha\|\beta\|_1 + (1 - \alpha)\|\beta\|_2^2 \leq s.$$

For a given data set, each penalty value λ corresponds to an effective constraint size s . A larger λ means stronger regularization and hence a smaller effective feasible region.

4.2 Choosing the Tuning Parameters

The tuning parameter λ should not be chosen arbitrarily. Elastic net also requires a choice of α . A standard procedure is K -fold cross-validation:

1. Specify candidate values of λ and, for elastic net, α .
2. For each candidate specification, fit the model on the estimation folds and evaluate prediction error on the held-out fold.

3. Average the held-out error across the K folds.
4. Select the candidate with the smallest cross-validation error and refit the model on the full estimation sample.

Regularization therefore separates two tasks:

- The **model parameters**, such as β , are estimated conditional on the tuning parameters.
- The **hyperparameters**, λ and, for elastic net, α , are chosen using predictive performance, commonly through cross-validation.

4.3 Main Takeaways

- OLS can become unstable when the number of predictors is large relative to the sample size or when predictors are strongly correlated.
- Ridge uses an ℓ_2 penalty and generally retains all predictors.
- The lasso uses an ℓ_1 penalty and can produce exact zeros, yielding a sparse model.
- Elastic net combines the two penalties and is useful when predictors are correlated but sparsity remains desirable.
- Regularization exchanges some bias for lower variance and improves prediction when the variance reduction exceeds the increase in squared bias.
- Cross-validation is commonly used to select the regularization strength and the elastic-net mixture.